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Smart vehicle management using cost effective approach

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ABSTRACT

With the high rise in climate conditions and thus the overcrowding of roads, carpooling is getting very popular when it involves eco-friendly and affordable travel options. Ridesharing is when more than one person shares a ride in their own vehicle. Carpooling minimizes pollution because a lesser number of cars are on the road. This is often also economical because the travel costs are divided between the drivers. The objective of this project is to develop a web application that provides a simple riding platform between the car owner and car user. This project enables users to access mobility assets own by others exactly when they need it. It shows a medium for available cars to pick up them on the interest of car owners with time and capacity. The web application is said to be common, which suggests that this can work for any passenger worldwide. Anyone participating in a tour can check-in at the rendezvous to admit the opposite people as soon as they arrive at the meeting point. Our web application is going to be having two types of users: First, the travelers (drivers) who drive their vehicle (currently cars only) to shuttle for work every day, and a few travelers got to share the value of fuel and other costs. Second, travelers (passengers) who also visit to work every day; besides they use conveyance or other means of transport that takes longer than usual to succeed in their terminus, making them feel drained and unclear at the top of the day. Our project is simply an effort to realize these two goals, the passengers and the drivers, building a relationship of faith for yourself and doing a little good for the surroundings as a responsible and informed person.
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1. Introduction

With the high rise in climate conditions and thus the overcrowding of roads, carpooling is getting very popular when it involves eco-friendly and affordable travel options. Ridesharing is when more than one person shares a ride in their own vehicle. Carpooling minimizes pollution because a lesser number of cars are on the road. This is often also economical because the travel costs are divided between the drivers.

Traveling by oneself can be traumatic. So, if you've got people with you on a tour, the strain can be reduced and there's also the opportunity to interact and have more fun on the tour. Locating carpools to share a tour with may be a challenge because it is difficult to seek out an individual who goes to an equivalent place as you at a given time. Many websites and applications are developed to assist people to share rides. With these approaches, users can make and share their trips as well as meet new individuals and have more fun [1].

The objective of this project is to develop a web application that provides a simple riding platform between the car owner and car user. This project enables users to access mobility assets own by others exactly when they need it. It shows a medium for available cars to pick up them on the interest of car owners with time and capacity. The web application is said to be common, which suggests that this can work for any passenger worldwide. Anyone participating in a tour can check-in at the rendezvous to admit the opposite people as soon as they arrive at the meeting point.

Our web application is going to be having two types of users: First, the travelers (drivers) who drive their vehicle (currently cars only) to shuttle for work every day, and a few travelers got to share the value of fuel and other costs. Few people may want a partner to go along with them throughout the tour. These persons may use our application to make their driver profiles in the "Offer a ride" section, including their path to terminus, the fare expected by passengers, the car, the number of passengers needed and the details of their time of

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departure from home off to terminus and when leaving the office. This helps other categories of users analyze their necessity and select drivers correspondingly. Most significantly, drivers are advised to provide their license numbers for security purposes.

Second, travelers (passengers) who also visit to work every day, besides they use conveyance or other means of transport that takes longer than usual to succeed in their terminus, making them feel drained and unclear at the top of the day. The commuters may use this application to seek out drivers whose starting point and terminus are an equivalent as them and to connect to the drivers and set tariffs, rendezvous and other required information's. The commuters are requested to also make their profile, doing so our application can filter the drivers according to their need. Our application also provides passengers with user ratings and filters accordingly, i.e. high and low ratings, so passengers can use our application and we can also meet their presupposition.

Our project is simply an effort to realize these two goals, the passengers and the drivers, building a relationship of faith for yourself and doing a little good for the surroundings as a responsible and informed person.

The web application into account, our carpooling application, may be a new idea. Which works as genesis of innovation for our task. All ongoing carpooling procedures are:

- Time absorbing.
- Requires tons of designing ahead.
- Requires multiple rounds of communication with the sort of a sequence of emails or a sequence of telephone calls.

Living in the metro cities like Delhi NCR we had observed that on a daily basis huge traffics jams are being faced by the public this mainly due to the large number of vehicles especially private cars. Also, we observed that the number of persons per car is on an average is either one or two. While there is heavy crowd in public transport so we got an idea to create a carpool web application so as to ease the travelling expenses as well as inconvenience caused to people due to traffic jams.

Another source of motivation for developing this web application is that of College, Gr. Noida. The teachers and students of this college have a problem that they have to fight for the elevator. That's why we come up with a carpool for comfort. Then we assumed; Why don't we start a web-based carpool application as a present to teachers and students? This really gave us the passion to leave after completing this project.

The Smart Vehicle Sharing as the name suggests, is an alternative to public transport. This is a website open to all citizens. Citizens can log into the site to avail carpool services. The person can either login as a driver or a passenger by simply registering themselves by filling the online registration form.

The passenger can only avail carpool services after he/she is logged in. The passengers can search for available cars as per the need. The driver can create a carpool which is shown as an availability to the passenger. The passenger can book a ride and after successful booking a confirmation SMS is sent on the mobile number of the passenger that he has mentioned while registering. The SMS contains the time, date, name and the contact number of the driver. The passenger can directly communicate with the driver via phone and give his/her location to him. Similar to the passenger, the driver also receives a SMS contain the details of the upcoming ride such as the contact, name of the passenger. The driver has the flexibility to either accept or reject the ride and update the corresponding status on the website.

Therefore, this project is only an attempt to achieve these two goals, the passengers and the drivers, to create a relationship of faith for yourself and to do a little good for the environment as an educated and responsible person.

2. Literature survey

The first step in our project was to evaluate the various carpooling applications available to determine requirements and improvements. We used their Google program to seek out carpooling web applications by entering the key "carpooling". A related web application was displayed in the results, which contains the same icon.

We found out the following from various websites which is shown in the below Table 1:

3. Problem statement

Rich people who may render a car every day do not have to think more than once before splurging it [2]. While ordinary person double think before splurging and have to save travel fee. In today's generation there is a big requirement of carpooling because of the continuous rise in fuel pricing and there is more than one place to travel in a day. Therefore, our carpooling service won't only guarantee you about safety your journey but also will offer you a reasonable ride anytime and anywhere in your area. In the Below Table 2, there are some statistics which were obtained when carpooling doesn't exist.

Table 1
Existing system.

WEBSITES	FEATURES	LIMITATION
BlaBlaCar	Find a ride, Create a ride, good verification	For intercity purpose only
Carpooling.in	Find a ride(regular/casual), create a ride(regular/casual)	Poor security
Co-Yatri	Find and create a ride	Lack of user interaction
PoolMyCar.in	Find and create a ride, verification through HR of company	Only for corporate people

Table 2
Statistical details.

Details	Amount Spent/Number of Cars
Average monthly Spending on a vehicle	4065 (Rs.)
Savings if carpooling with 1 person	2033 (Rs.)
Saving if carpooling with 2 persons	2710 (Rs.)
Saving if carpooling with 3 persons	3049 (Rs.)
Number of cars reduced on road if carpooling with 1 person	181781 cars
Number of cars reduced on road if carpooling with 2 persons	70478 cars
Number of cars reduced on road if carpooling with 3 persons	148,906 cars
Amount Saved on petrol per year	1310.98 crore (Rs.)

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4. Services behind the project

The various services behind the project are as follows:

4.1. Java

Java may be a general-purpose programing language that's class-based, object-oriented and specially designed so that there are a few implementation dependencies as possible. Application developers should be ready to write once and execute anywhere (WORA). This suggests that compiled Java code can run on all platforms that support Java without the necessity for recompilation. Java applications are usually compiled into byte code which will run on any Java Virtual Machine (JVM) no matter the underlying computer architecture [8]. Java syntax is analogous to C and C++, but has fewer low-level functions than both. Java is currently one among the foremost popular programming languages, especially for client–server web applications [8].

4.2. Javascript

JavaScript, often abbreviated as JS, maybe a higher-level interpreted programing language. It's a language that's also characterized as dynamic, weakly typed, prototype-based and multi-paradigmatic. JavaScript is executed on the client-side of the online, with which the behavior of the online pages is often designed/programmed when an occasion occurs. JavaScript is straightforward to find out and a strong scripting language that's often wont to control website behavior. Alongside HTML and CSS, JavaScript is one of the core technologies of the planet Wide Web. JavaScript enables interactive websites and is an integral part of web applications. The overwhelming majority of internet sites use it, and enormous web browsers have a fanatical JavaScript engine to run it.

4.3. Html

Hypertext Markup Language (HTML) is the standard markup language for creating websites and web applications. Cascading Style Sheets (CSS) and JavaScript form a triad of cornerstone technologies for the World Wide Web. Web browsers receive HTML documents from a web server or local storage and render the documents in multimedia web pages. HTML describes the structure of a website semantically and originally contains information about the appearance of the document.

4.4. Java Server Pages

Java Server Pages (JSP) is a technology that software developers can use to create dynamically generated web pages based on HTML, XML, or other types of documents. JSP was released in 1999 by Sun Microsystems [1]. JSP is similar to PHP and ASP, but uses the Java programming language [7]. A compatible web server with a servlet container such as Apache Tomcat or Jetty is required to deploy and run Java Server Pages.

4.5. Css

Cascading Style Sheets (CSS) is a style sheet language that describes the presentation of a document written in a mark-up language such as HTML. Alongside HTML and JavaScript, CSS is a cornerstone technology of the World Wide Web.

CSS was developed to allow the separation of presentation and content, including layout, colors and fonts. This separation can improve the accessibility of content, offer more flexibility and control when specifying presentation properties, enable the joint formatting of multiple websites by specifying the relevant CSS in a separate CSS file, and reduce the complexity and repetition of the structural content.

4.6. Mysql

MYSQL is an open source relational database management system (RDBMS). His name is a combination of "My", the name of the daughter of co-founder Michael Wideners, and "SQL", the abbreviation for Structured Query Language.

MySQL is free and open source software under the terms of the GNU General Public License and is also available under a variety of proprietary licenses. MySQL was owned and sponsored by the Swedish company MySQL AB, which was bought by Sun Microsystems (now Oracle Corporation). In 2010, when Oracle acquired Sun, Wideners abandoned the open source MySQL project to create Maria DB.

4.7. Sms API (WAY2SMS)

An application programming interface (API) consists of a number of subroutine definitions, communication protocols and tools for creating software. In general, it is a series of clearly defined communication methods between different components. A good API makes it easier to develop a computer program by providing all the building blocks that are then put together by the programmer.

An API can be for a web-based system, operating system, database system, computer hardware, or software library. An API specification can take many forms, but often contains specifications for routines, data structures, object classes, variables, or remote calls. POSIX, Windows API and ASPI are examples of different types of APIs. Documentation for the API is usually provided to facilitate use and implementation. We have implemented SMS APIs (way2sms) in our project.

4.8. Eclipse IDE

Eclipse is an integrated development environment (IDE) that is used in computer programming and is the most commonly used Java IDE. It contains a basic work area and an expandable plug-in system to adapt the environment. Eclipse is mainly written in Java and is mainly used to develop Java applications. However, it can also be used to develop applications in other programming languages via plug-ins, including Ada, ABAP, C, C ++, C #, Clojure, COBOL, D., Erlang, Fortran, Groovy, Haskell, JavaScript, Julia , Lasso, Lua, NATURAL, Perl, PHP, Prolog, Python, R, Ruby (including Ruby on Rails framework), Rust, Scala and Schema. It can also be used to develop documents with LaTeX (via a TeXlipse plug-in) and packages for the Mathematica software. The development environments include the Eclipse Java Development Tools (JDT) for Java and Scala, Eclipse CDT for C / C ++ and Eclipse PDT for PHP.

4.9. Internet connectivity

The internet connection is important to access the web application and web-based content of our project. A website or website is a collection of related network resources, such as websites and multimedia content, which are usually identified by a common domain name and published on at least one web server.

4.10. Apache Tomcat Server

Apache Tomcat, often referred to as Tomcat Server, is an open source Java servlet container developed by the Apache Software

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Foundation (ASF). Tomcat implements several Java EE specifications, including Java Servlet, Java Server Pages (JSP), Java EL, and Web Socket, and provides an HTTP "pure Java" web server environment that can run Java code.

Developed and maintained by an open community of developers under the auspices of the Apache Software Foundation, which was released under the Apache License 2.0 license, Tomcat is open source software

5. Specific requirements

The various requirements for developing the projects are as follows:

5.1. External interface requirement

The user interface must be understandable and easy to use. Also, it should be interactive. The system should have features so that it asks the user and administrator to log in and have some rule to enter correct input. External interface requirements like hardware, software, or database elements to which a system or component must be connected. This portion contains all the needed details which guarantees that the system is properly communicating with external parts. If different parts of the product have different external interfaces, insert an instance of this section into the detailed requirements for each of these parts.

5.2. User interface

The logical properties of interfaces between user and the product is described by user interface. This may include sample screens, any gui standards to follow or family style guidelines, screen layout restrictions, standard buttons and features (e.g. help) that appear on each screen, keyboard shortcuts, standards for displaying error messages, and soon. Define the software components for which a user interface is required. User interface design details should be documented in a separate user interface specification.

The software provides a good graphical interface for the user to control the system and execute the demand task with the following functions:

- Simplicity and ease of travelling.
- Easy to add a ride.
- Easy to search a ride.
- Feedback

5.3. Functional requirements

Functional requirements are those that relate to the technical functionality of the system. The functional requirements are classified as follows:

5.3.1. Hardware requirements

The hardware requirements report properties (logical and physical) between the software product and the hardware components of the system of each interface. This can involve the supported device types, data type and interactions between the software and the hardware, and also the protocols (communication) that is used.

The various hardware requirements are as follows:

- **Operating System:** Windows 7
- **Hard Disk:** 20 GB
- **RAM:** 128 MB
- **Processor:** Pentium III 630 MHz
- **Monitor:** Colour monitor

5.3.2. Software requirements

The software requirements describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries and integrated commercial components. Identify the data items or messages entering and leaving the system and describe the purpose of each item. Describe the services required and the type of communication. See documents describing detailed protocols of the application programming interface [3]. Identify data that is shared by software components. If the data sharing mechanism needs to be implemented in a certain way (e.g. using a global data area in a multitasking operating system), specify this as an implementation restriction.

The various software requirements for our projects are as follows:

- **Database:** MySQL Workbench
- **Browser:** Any of Mozilla, chrome, IE etc.
- **Web Server:** Tomcat 9.0
- **Software Development Kit:** Java jdk 1.7 or above
- **Scripting Language:** Java Server Pages
- **Database JDBC Driver:** MySQL database connector (Version 6.0.3 and above)

5.3.3. Performance requirements

There are static and dynamic numerical needs for the system or for human interaction with the system. Static numerical needs can involve the number of terminals to be supported, the number of users to be supported at an equivalent time and therefore the amount and sort of data to be processed [4]. Dynamic numerical requirements can include the amount of transactions and tasks, also because the amount of knowledge to be processed within a particular period of your time for both normal and peak utilization conditions. The performance of the pc depends on the performance of the software. The software can make as many entries as long because the database size is large enough. This is able to depend upon the space available.

6. Design and implementation

6.1. Use case

A use case diagram may be a graphic representation of the interactions between the weather of a system. A use case may be a method utilized in system analysis to spot, clarify, and organize system requirements. During this context, the term "system" refers to something that's being developed or operated, for instance an internet site for the sale and repair of shipping products. Use case diagrams are utilized in UML (Unified Modelling Language), a typical notation for modelling real objects and systems.

Use case diagrams contains actors, use cases and their relationships. The diagram is employed to model the system/subsystem of an application. A case-by-case diagram captures a selected functionality of a system. Variety of use case diagrams are therefore wont to model the whole system.

6.1.1. Passenger module

The Passenger Use Case is demonstrated in the Fig. 1.

6.1.2. Driver module

The Fig. 2 explains the Driver use case.

6.1.3. Combined use case

The combined use case diagram is shown in Fig. 3.

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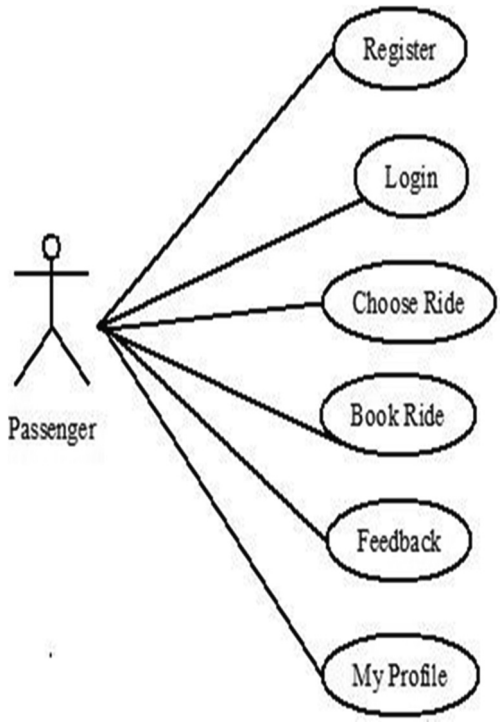


Fig. 1. Passenger use case.

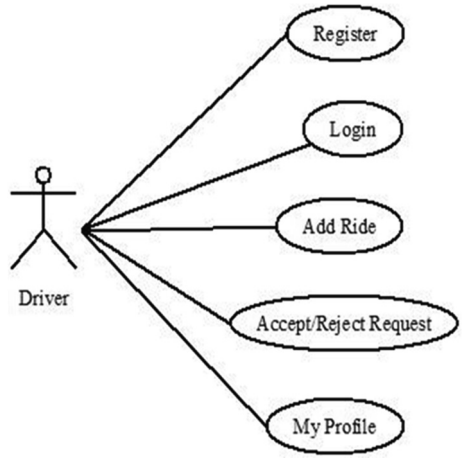


Fig. 2. Driver use case.

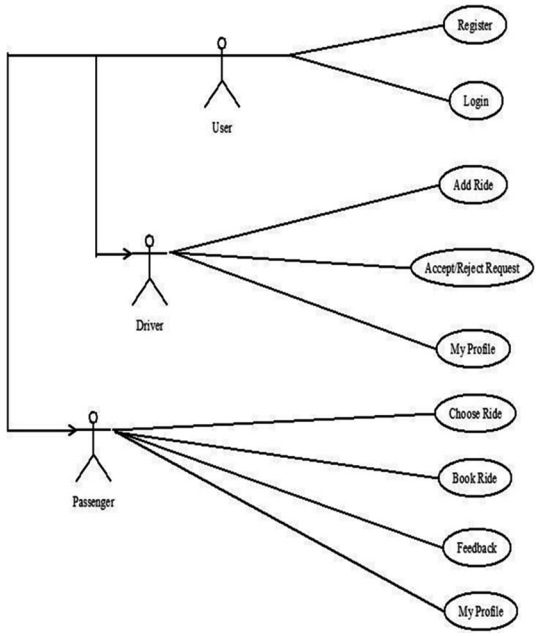


Fig. 3. Combined use case.

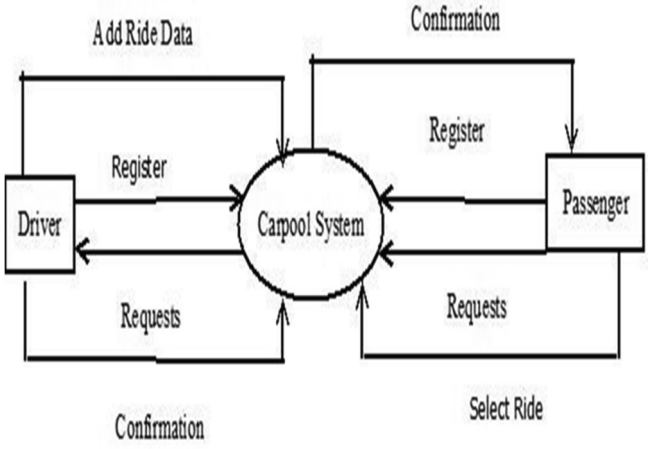


Fig 4. Level 0 DFD.

6.1.4. Data flow diagram

A process flow diagram in an information system is a graphic representation of the data flow. It is able to display the data flow for incoming data, outgoing data and also for stored data. The DFD notifies nothing about how data goes by the system [5]. A data flow diagram (DFD) is a way to show a data flow of a process or a system (usually an information system). The details related to input and output of process and entity is given by DFD. A data flow diagram does not involve control flow, decision rules and loops. Certain operations based on the data can be represented by a flow chart. The level 0 DFD, Level 1 DFD, Level 2 DFD is explained in the Fig. 4, Fig. 5 and Fig. 6 respectively.

6.1.5. Entity relation diagram

An entity relationship model (or ER model) distinguish interdependent interesting things in a certain area of knowledge. A basic ER model contains of entity types (which classify things of interest) and specifies relationships which will exist between entities (instances of those entity types). An entity relationship model is usually the results of systematic analysis to define and describe what's important for processes during a business area. It doesn't tell about business processes; just one business data schema is there in graphical form [6]. It is usually represented in graphical form as boxes (entities) associated by lines (relationships) that shows the associations and dependencies between entities. The ER diagram is shown in the below Fig. 7.

7. Testing

Software testing is an inquiry that is accomplished to give stakeholders with details related about the quality of the software product or service being tested [1]. Software testing can also furnished an objective, independent view of the software so that the company can evaluated and understand the risks of software implementation. Test techniques involve running a program or

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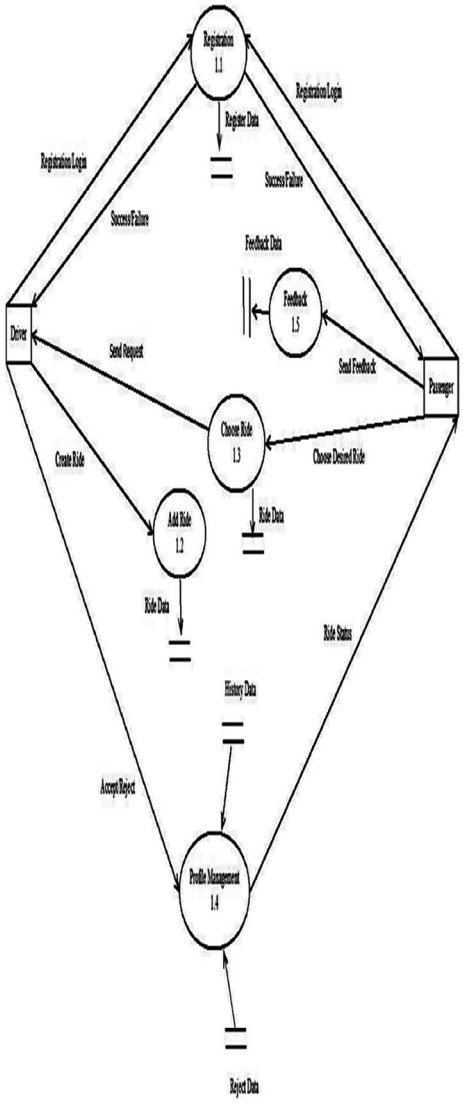


Fig 5. Level 1 DFD.

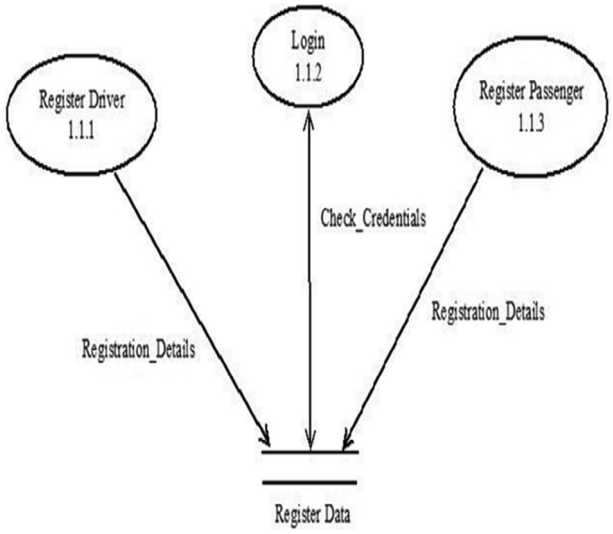


Fig. 6. Level 2 DFD.

application with the motive of searching software errors (bugs or other defects) and examine that the software product is proper for use.

7.1. Test cases and results

A test case is a rule set or variables under which a tester finds determines whether a system under test encounters the needed or is working properly. The process of developing test cases can also find problems with the requirements or design of an application.

8. Module description

In this section, we have provided information of the various modules which are implemented in our website. In this project, the driver and passenger can interact with each other via SMS for booking and confirming the ride after using with the website.

This helps the concept of carpool to be easily implemented and provide benefit to both the ends. This Project has been divided into the following modules:

8.1. Registration module

In this module, two types of users can register in our website:

- **Register as a passenger**
- **Register as a driver**
- **Login**

Register as a Passenger: In, register as a passenger a person needs to fill the required details such as name, address, mobile no, Aadhar no, license no, etc. Also, the passenger needs to Agree Terms and Conditions of our website in order to register themselves successfully.

Register as a Driver: In, register as a driver, a person needs to fill a form containing his name, mobile no. address, Aadhar no, license no. vehicle no, email id etc. Further the driver needs to agree Terms and Conditions of our website in order to register themselves successfully.

Login: In this module, user needs to enter login credentials which contains email id and password. Also, the user needs to select the type of user i.e. passenger or driver.

8.2. Add ride module

This module is accessible only to drivers. Here the driver can add a ride as per his availability. The driver can enter the following details:

- **Travelling from:** Enter the city and area from where the driver starts his journey.
- **Travelling to:** Enter the city and area of the destination of the journey.
- **Seats:** The number of seats he wants to make it available for the carpool.
- **Date:** Here, the driver selects the date of journey.
- **Time**: Driver enters the time of journey.
- **Fare:** Driver enters the fare of each person per ride.

Before proceeding to submit the form, the user can see a fare chart which indicates the charges to be taken from passengers. It is just our effort to reduce the rate of charges incurred from passengers. However, if driver and passenger agree to a common travel expense then it is up to them. Also, if the driver is taking

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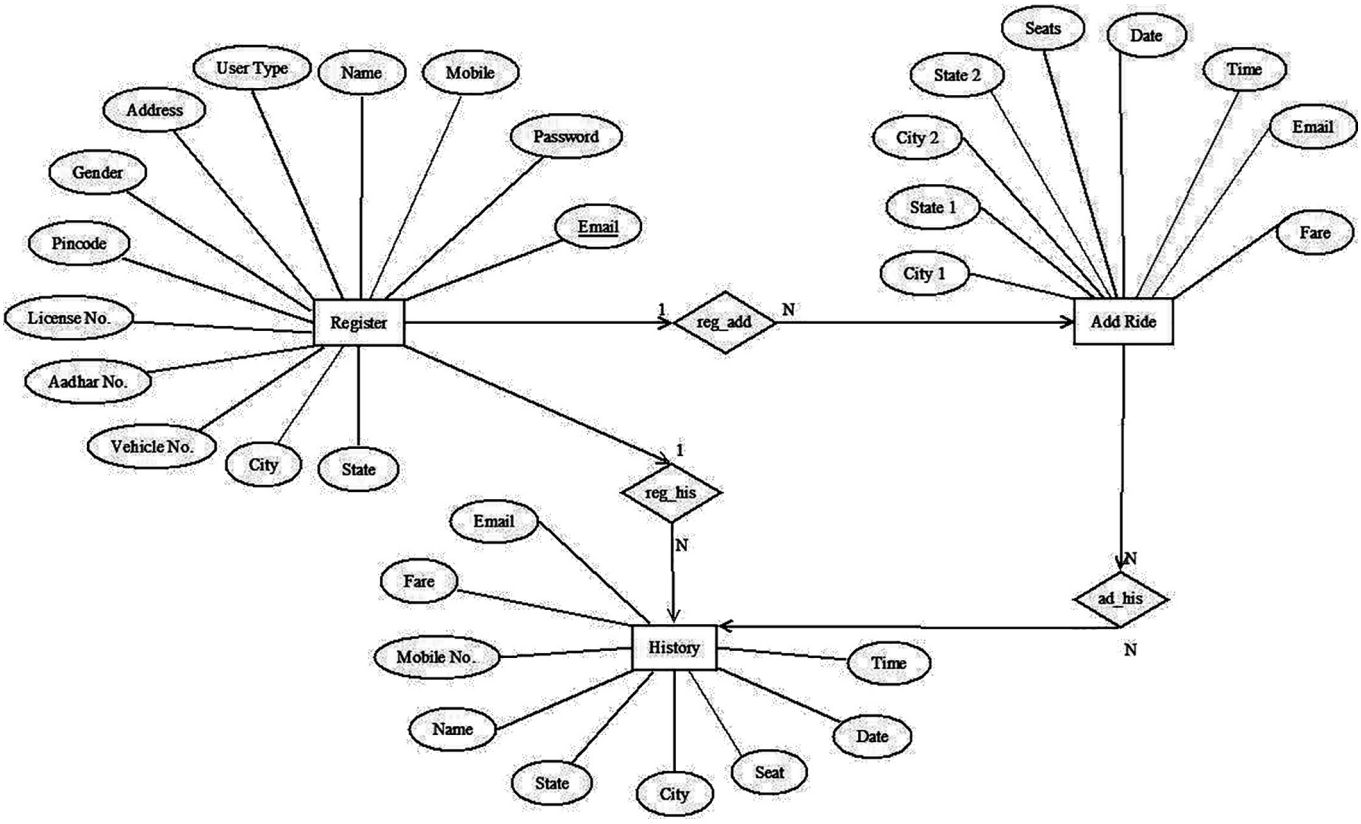


Fig. 7. ER diagram.

excessive charges then the passenger can complain against him in our website through the information contained in contact tab.

8.3. Choose ride module

In this module, the passenger needs to first login and can find his respective ride (if available). The passenger needs to enter the following fields:

- **Travelling from:** city, area.
- **Travelling to:** city, area.
- **Date of ride:** The passenger can select the desired date.

The entered details are matched with the entries of ride table in the database and the matched tuples are fetched and displayed on the client's side with the following details such as Name, mobile no, source city, source area, destination city, destination area, seats, date, time, fare and a button to book ride.

As soon as the passenger clicks on book a ride, a SMS is sent to the corresponding driver and a message is displayed on the screen to wait for the response of the driver. The below figures depict the choose ride module.

8.4. My profile module

This module is divided mainly into two parts:

My Profile passenger: In this module, the details of the passenger is shown as soon as the passenger gets log in to the website such as name, email id, mobile no, address, Aadhar no. This module also consists of the history of the **rides taken by** the passenger till date with the driver's name, travelling from (city, area), travelling to (city, area), date and time.

This allows the passenger to have a look at his previous journeys and find the driver's details whom he wants to continue carpool in future. Figure 7.4.1 depicts the passenger profile.

My Profile driver: In this module, the details of the passenger is shown as soon as the passenger gets log in to the website such as name, email id, mobile no, address, Aadhar no, license no and vehicle no. This module also consists of the requests send by the passenger to the driver where the driver has the choice to either accept or reject the ride. If the driver accepts the ride, the requesting passenger gets a confirmation message for the ride along with the driver's name and contact no.

However, if the driver chooses to reject the ride for the requesting passenger, a SMS is sent to the passenger regarding rejection of ride from driver's end and to look for more rides in the website. Figure 7.4.2 depicts the driver profile.

9. Feedback module

Once the passenger successfully completes the ride, he/she can give rating to the driver on the basis of his experience during the ride. This will allow us to know more about each driver in our website registered. This also helps the user to select more appropriate driver as per the feedback.

10. Conclusion

The carpooling system is a very effective way to reduce pollution and congestion of vehicles in cities. It also offers an environmentally friendly way to travel. It also offers the opportunity to meet new people. As today, most people prefer private vehicles due to delays caused by public transport systems and the luxury of private vehicles.

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Pre-registration ensures security, as only identified people enter the vehicle to build trust. The registered persons can be assigned specific days on which they should take their private vehicle, so that the registered passengers are not inconvenienced by the daily commuting. Thus, the proposed carpooling system will effectively reduce pollution.

The project is made purely for fulfilling the purpose of traveling within the city extremes. The project is benefit for the people who prefer frequent traveling. With the perspective of making use of the daily activity of common people to help and earn easy money, with not much effort. No more hustling for the vehicles. We work for saving the high transportation charges and increase the availability of the vehicles at the areas where we have trouble in transportation.

This project, Smart Vehicle Sharing is a project which has been designed with the perspective of a start-up and is in its beginner stage. And as it would expand its services over the time, it would make the motive much easier to achieve. The technology used can easily be handled and hence it provides a better support for future maintenance.

We would be developing a full-fledged android application in future.

CRediT authorship contribution statement

Muneesh Chandra Trivedi: Conceptualization, Data curation, Supervision, Validation. **Paritosh Tripathi:** Writing - original draft, Writing - review & editing. **Vineet Kumar Singh:** Formal analysis, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Further Reading

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